ANSI/IES LM-86-20(R2023)

Illuminating
ENGINEERING SOCIETY

APPROVED METHOD: MEASURING LUMINOUS FLUX AND COLOR MAINTENANCE OF REMOTE PHOSPHOR COMPONENTS

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has been approved by IES.
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should be directed to IES.

Prepared by
The IES Testing Procedures Committe

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1.0 Introduction and Scope

1.1 Introduction

Remote phosphor is an emerging technology in LED lighting, which to date has not yet been covered by light measurement standards.

Due to the gradual nature of decay mechanisms, a remote phosphor component may undergo shifts in spectral emission, color rendering ability, and luminous efficacy that degrade performance over time. Thus, an LED based lamp or luminaire with a remote phosphor component could experience shifts in performance outside the product's specification over the product's lifetime.

With prolonged exposure to high irradiance levels from pump LED sources, the remote phosphor component may exhibit changes in its chromaticity, correlated color temperature (CCT), and color rendering index (CRI). The causes of the changes may be due to the types and quantities of phosphors used in the remote phosphor component. More specifically, current practice indicates that LED lamps or luminaires with lower CCT (e.g., 2700 K) and higher CRI (e.g., 90) have slightly more-rapid degradation characteristics.

This Lighting Measurement (LM) document addresses the test method for measuring degradation behavior of the remote phosphor component. In addition to using the method of testing an entire remote-phosphor LED lamp or luminaire per *ANSI/IES LM-84-20, Approved Method: Measuring Optical Radiation Maintenance of LED Lamps, Light Engines, and Luminaires*,[1] this LM provides an alternative method, whereby the separable remote phosphor component can be tested.

For the purposes of this document, it is assumed that degradation characteristics of the remote phosphor components do not depend on pump device design or pump source layout, and that degradation is characterized entirely by the temperature and associated irradiance level of the remote phosphor. Maintenance impact due to pump source wavelength can be evaluated through use of pump sources at the extremes of the remote phosphor dominant wavelength range; however, this document does not specify the pump source wavelength. Therefore, the remote phosphor components tested per this LM can be any pump source that generates similar or lower irradiance levels at the same or lower maximum and average temperatures on the remote phosphor components. The pump devices or sources may be in the form of discrete devices, chip-on-board configurations, arrays, or any other configuration.

In *ANSI/IES LM-80-20, Approved Method: Measuring Luminous Flux and Color Maintenance of LED Packages, Arrays, and Modules* (see **Section 2.2**), the luminous flux and color maintenance tests are defined based on LED case temperature and LED forward current. In this document, the luminous flux and color maintenance tests are defined based on the remote phosphor component's maximum and average temperatures and irradiance level. As the forward current increases LED case temperature, similarly the irradiance level increases the operating temperature of the remote phosphor component.

1.2 Scope

This Lighting Measurement (LM) document provides the method for measurement of luminous flux and color maintenance of remote phosphor components. The method describes the procedures to be followed and the precautions to be observed in order to obtain uniform and reproducible luminous flux and color maintenance measurements under standard operating conditions.

This document does not cover the determination of the performance rating of products, for which individual variations among the products should be considered. This Approved Method does not provide guidance or make any recommendation regarding sampling, nor regarding predictive estimation or extrapolation for luminous flux or chromaticity maintenance determined from actual photometric measurements.

2.0 Normative References

2.1 ANSI/IES LS-1-20

Illuminating Engineering Society. Lighting Science: Nomenclature and Definitions for Illumination

Engineering. New York: IES; 2020. Online: https://www.ies.org/standards/definitions/. (Accessed 2020 Jan 21).

2.2 ANSI/IES LM-80-20

Illuminating Engineering Society. Approved Method: Measuring Luminous Flux and Color Maintenance of LED Packages, Arrays, and Modules. New York: IES; 2020.

3.0 Definitions

3.1 air temperature (T_A)

The temperature of the air surrounding the DUT.

3.2 average irradiance on DUT ($E_{ave,DUT}$)

The total radiant flux $\Phi_{e,inc}$ from the pump LED source incident onto the remote phosphor component, divided by the remote phosphor component surface area A_{DUT} irradiated by the pump LED:

$$E_{ave,DUT} = \frac{\Phi_{e,inc}}{A_{DUT}}$$

The excitation beam diameter defining the area exposed on a two-dimensional DUT shall be defined by the 10-percent points of the peak irradiance for cases where the DUT is not uniformly exposed to pump LED source radiant flux.

For complex three-dimensional surfaces, the surface area can be calculated using the solid model file for the DUT. If the excitation surface is not uniformly exposed, the 10-percent points can be obtained through luminance measurement of the DUT emission surface or by using the pump LED source intensity distribution.

3.3 average surface temperature ($T_{s,avg}$)

The average temperature on the excitation surface of the DUT defined by its clear aperture.

3.4 conversion efficacy (K_{conv})

The ratio of the total luminous flux $\Phi_{v,emit}$ emitted from the emission surface of a remote phosphor component, to the total radiant flux $\Phi_{e,inc}$ incident on the excitation surface of a remote phosphor component at a specified wavelength characteristic, noted as K_{conv}, with units of lumens per watt (lm/W):

$$K_{conv} = \frac{\Phi_{v,emit}}{\Phi_{e,inc}}$$

The dominant, peak, or centroid wavelength of the pump source shall be noted. The luminous flux typically includes unconverted light from the pump LED source(s).

Additional details on conversion efficacy may be found in **Annex A**.

3.5 conversion efficacy maintenance

The conversion efficacy over time, expressed as a percentage of the initial (at time equal to zero) conversion efficacy at any selected total elapsed operating time.

3.6 device under test (DUT)

A remote phosphor component undergoing a maintenance test.

3.7 emission surface and clear aperture

Emission surface is the surface of a remote phosphor component from which photons emanate due to excitation of the phosphor by a pump light source, and from which some transmission of pump light is emitted. (Note that for remote phosphor components in which the phosphor is applied to a reflective material [diffuse or specular], the emission surface and the excitation surface may be coincident.)

Clear aperture is used in this document to designate the remote phosphor opening or surface region through which light can travel. For example, if a 60-millimeter round remote phosphor component is held by a housing with an opening of 55 millimeters, then the clear aperture of the remote phosphor is 55 millimeters.

3.8 excitation surface

The surface of a remote phosphor component irradiated by the pump LED source(s).

3.9 failure of DUT

A catastrophic failure that leads to the DUT no longer being able to function in the intended light source test assembly. Failure modes include but are not limited to the DUT melting, bending, or cracking.

3.10 maintenance pump LED source

LED irradiance source with or without mixing chamber that is used in conjunction with the DUT during the maintenance test and can be used for photometric measurement, assuming stable output performance per **Section 6.2**.

3.11 maintenance test

The operation test for the DUT when it is energized by the appropriate pump LED source under specific electrical and environmental conditions.

3.12 maximum surface temperature ($T_{s,max}$)

The highest temperature on the excitation surface of the DUT.

3.13 measurement interval

The elapsed time between subsequent photometric or subsequent electrical measurements.

3.14 reference pump LED source

The pump LED source with the same characteristics defined in **Section 3.10** that is not used during the maintenance test. It is used in conjunction with the DUT for photometric measurement only.

3.15 remote phosphor

A configuration implemented in an LED or LED lighting product—e.g., LED lamp, LED light engine, or LED luminaire—using a phosphor not in direct contact with the pump LED die.

3.16 remote phosphor component

An element or component that is mechanically separable from the pump LED source and contains one or more phosphor materials. The element is used in conjunction with LED packages, arrays, or modules to construct LED lamps, LED light engines, and LED luminaires. Such components may be of any shape. The phosphor material may be either applied to one or more surfaces of a transmissive and/or reflective material, or embedded in the material. For the purposes of this LM document, *mechanically separable* means that there is a gas or other suitable material between the LED die and the remote phosphor.

4.0 Pump LED Sources and Electrical Requirements

4.1 Pump LED Sources

4.1.1 Pump LED Source Characteristics. The characteristics of the maintenance and reference pump LED sources shall be obtained and recorded. The information for these pump LEDs should include at a minimum:

- Total radiant flux $\Phi_{e,inc}$ onto the remote phosphor component, in watts
- Dominant (if applicable), peak, or centroid wavelength
- Case temperature: the temperature of the thermocouple attachment point of the pump LED package, as defined by the manufacturer of the LED package

4.1.2 Measurement of Maintenance and Reference Pump LED Source Characteristics. Radiant flux and the related wavelength shall be measured per applicable standard procedures (e.g., ANSI/IES LM-85-20[2]).

LED package case temperatures should be verified with a thermocouple measurement system that complies with ASTM E230 Table 1 "special limits" (≤1.1 °C or 0.4%, whichever is greater).[3] A heat sink meeting the recommendations of the LED manufacturer should be used to meet the manufacturer's recommended case temperature.

Characteristics of the pump LED source(s) shall be measured and recorded at every photometric measurement interval. It is recommended that the reference pump LED source be used for all photometric measurements, since it will have minimal output change over time.

4.2 Power Supply Electric Requirements

The electronic power supplies used to energize the LED pump sources for maintenance test and photometric measurement should meet all of the requirements of the LED manufacturer. Of greatest concern is making sure to operate the LEDs within their specified current range. Where practical, it is recommended to under drive the devices so that radiant power maintenance is optimized. In general, non modulated constant current drivers are recommended to simplify photometric measurement.

5.0 Physical and Environmental Conditions for Maintenance Test

5.1 Temperature

5.1.1 Temperature on DUT Surfaces. DUT emission surface (viewable to measurement equipment) temperature shall be evaluated using an IR camera or IR thermometer or thermocouple. The highest temperature point on the irradiance incidence surface $T_{s,max}$ shall be identified and recorded. $T_{s,max}$ is best located on the DUT surface using an IR camera, but quantitative measurement of $T_{s,max}$ is best done with a thermocouple, because of uncertainties in the emissivity of the remote phosphor component.

A thermocouple shall be used to monitor the irradiance incidence surface temperature. A thermocouple measurement system complying with ASTM E230 Table 1 "Special Limits" (≤1.1°C or 0.4 percent, whichever is greater) shall be used to monitor the DUT irradiance incidence surface temperature $T_{s,max}$.[3] The surface temperature shall be monitored during the maintenance test. $T_{s,max}$ can be measured per the recommended procedures in **Annex B**.

5.1.2 Air Temperature. Air temperature surrounding the DUT shall be monitored and recorded. The location of the test thermocouple is chosen to best represent the surrounding air without introducing inaccuracies (e.g., due to light absorption). The thermocouple locations shall be reported. The temperature of the surrounding air shall be maintained at a temperature ≥25 °C unless otherwise specified. Higher ambient air temperatures (T_a) will require lower pump irradiance levels for a given maximum DUT operating temperature, and vice versa

In some cases, selection of several ambient air temperature conditions may be chosen by the DUT manufacturer.

5.2 Air Flow

Natural convective air movement is allowed. Air shall not be circulated artificially.

5.3 Humidity

During the maintenance test, relative humidity (RH) shall be maintained to less than 65%, or to another predefined RH level, within a tolerance of ±5%, throughout the maintenance test. The chosen relative humidity test condition (less than 65% or predefined) shall be reported.

6.0 Photometric Measurements

6.1 Pump LED Source Measurement

Reference pump LED sources or maintenance pump LED sources shall be used for the photometric measurements. Driving current for reference pump LED sources or maintenance pump LED sources shall be recorded at each photometric measurement interval.

6.2 Pump LED Source Radiant Flux Maintenance

If the radiant flux of the maintenance pump LED sources or reference pump LED sources measured deviates by 10% or more from the original value at 0 hours, the pump LED sources shall be adjusted (e.g., forward current) or replaced at the measurement interval when the variation is identified. The pump LED source adjustment or replacement shall be reported. Where the maintenance pump sources are not used for photometric measurements of the DUTs, it is recommended that they be verified at each DUT measurement interval.

6.3 Measurements of Photometric and Color Characteristics

For photometric measurements, the DUT shall be operated with the maintenance or reference pump LED source(s). Total luminous flux and chromaticity shall be measured in conformance with the appropriate method for the LED light source under test. The ambient temperature during the measurements shall be set to 25 °C ± 1 °C.

6.4 Conversion Efficacy Calculation

The conversion efficacy shall be calculated per **Section 3.4**.

7.0 Maintenance Test Procedures

7.1 Seasoning or Aging

No seasoning or aging of the DUTs is required prior to the maintenance test.

7.2 DUT Exposure to Pump LED Sources (Irradiance)

The DUT shall be exposed to a specified average irradiance level (see **Section 3.2**) throughout the maintenance test. The maintenance pump LED sources(s) and reference pump LED source(s) shall be operated within the manufacturer's recommended limits

7.3 DUT Maximum Surface Temperature

The DUT maximum surface temperature $T_{s,max}$ shall be located and set to the specified value within ±5 °C variation at time zero. This location shall then be monitored at every test interval. It is recommended that at least one maximum surface temperature location be used per maintenance test.

7.4 DUT Average Surface Temperature

The DUT average surface temperature $T_{s,avg}$ over the excitation surface shall be obtained and recorded at each test interval.

7.5 Test Duration, Measurement Interval, and Time Keeping

Initial luminous flux, chromaticity (CIE *x,y* or *u',v'* values), spectral power distribution, and conversion efficacy measurements shall be performed prior to the starting of the maintenance test and recorded as the 0 hours data.

Photometric measurements of the DUT shall be made at each measurement interval of the maintenance test duration.

The maintenance test duration and measurement intervals shall be based on and consistent with the design operating life; the intent of the test, including evidence of compliance to the regulations requested; and planned analysis of the data (e.g., data for predictive luminous flux maintenance modeling, or data for actual luminous flux maintenance validation).

8.0 Test Report

The report shall list all pertinent data concerning the test conditions, type of equipment, and types of DUTs. The applicable items in this section shall be recorded at each measurement interval, if there is a change from the previous measurement. Typical items reported are:

- Administrative Information
  - Testing agency identification
  - Report issue date
  - Testing start date
  - Testing completion date
  - Individual(s) performing the testing
  - Individual(s) reviewing and approving the test results
- DUT Identification
  - DUT manufacturer's name
  - DUT identification, e.g., model number
  - Description of DUT, including nominal CCT, CRI, size, shape, thickness, and substrate
  - Sampling method and number of DUTs (sample size) tested
- Test Conditions and Pump LED Sources
  - Average temperature, $T_{s,avg}$, of DUT irradiance incidence surface, measured over the clear aperture (not between the 10-percent points of the illuminance profile)
  - Maximum temperature, $T_{s,max}$, of DUT irradiance incidence surface
  - Ambient temperature, T_a
  - Relative humidity (RH) level
  - Mixing chamber geometry and materials (if applicable)
  - Description of pump LED sources including brand name and part number, number of LEDs, physical geometry, and optics

  - Forward current and voltage for pump LED sources used during maintenance test and for reference pump LED source used at photometric measurement
  - Radiant flux for pump LED sources used during maintenance test and for reference pump LED source, along with the associated peak, dominant, or centroid wavelength characteristic
  - Average irradiance on DUT, $E_{\mathrm{ave,DUT}}$
- Test Equipment
  - Description of testing equipment.
  - Auxiliary equipment used for test.
- Maintenance Test Duration (in Hours)
- Measurement Intervals (in Hours)
  - Failed DUTs
  - Number of DUT failures
  - Approximate time each failure was observed
  - Pump LED source adjustment or replacement
  - Pump LED source failures
- Results
  - Luminous flux, $\Phi_{\mathrm{v,emit}}$
  - Chromaticity (CIE *x*,*y* or *u*′,*v*′ values)
  - Spectral power distribution
  - Conversion efficacy, K_{conv}
- Optional Items
  - Forward current, voltage, and radiant flux for: each DUT; pump LED sources used during the maintenance test; and reference pump LED source used to measure the photometry and chromaticity of the DUT
  - Statement of uncertainties (if required).

Annex A – Conversion Efficacy of Remote Phosphor Components

The remote phosphor material luminous efficacy can be expressed in terms of a *conversion efficacy*, which is the ratio of lumens converted to the flux incident upon the remote phosphor material. As an example, if a remote phosphor component emits 200 lumens of total luminous flux and the incident excitation power from the pump LEDs ($\Phi_{\mathrm{e,inc}}$) is 1 watt, then the conversion efficacy is 200 lumens per radiant watt, noted as 200 lm/$\Phi_{\mathrm{e,inc}}$. Conversion efficacy is a function of the pump source spectral energy distribution and is normally measured at a specific peak or dominant wavelength with a standard test source geometry.

In this document, conversion efficacy includes down-converted and transmitted radiation, which comprise the total luminous flux measured. At a system level (combining pump LEDs and remote phosphors), the system efficiency will also include the wall plug efficiency of the pump LED sources, the efficiency of any secondary optics, such as reflectors or diffusers, and any light lost due to the reflective material between the pump LEDs and the remote phosphor. This reflective material, which may be diffuse or specular, is referred to as the *mixing chamber*; its purpose is to recycle the approximately 50 percent of converted light that does not escape the system on a first pass. It also mixes the light to provide a more uniform color characteristic.

Although the term *mixing chamber* is used here, in some cases it may simply be the planar reflective material or reflective coating on the LED PCB. The mixing chamber “volume” then is composed partly of the remote phosphor material rather than a three-dimensional reflector.

The absolute conversion efficacy of a remote phosphor system will be partly determined by the mixing chamber efficiency. However, in this document the relative measure of conversion efficacy and the maintenance of the relative conversion efficiency are critical. It is recommended to use reflective mixing-chamber materials that are UL and/or ASTM rated, and the maximum operating temperature of the material should not be exceeded for the duration of the test.

It is recommended to use mixing-chamber materials that have been tested for weathering per QUV accelerated weathering testing or equivalent, per ASTM G154, ASTM D4329, ASTM D4587, or ISO 4892. The QUV accelerated testing is far beyond what these materials will experience in actual application, since the peak wavelength is 340 nm (peak wavelengths for remote phosphor LED pumps are typically no less than 445 nm). A typical irradiance level for QUV accelerated weathering for UVA-340 is

0.68W/m². The mixing chamber material should also have a maximum operating temperature that is not exceeded in these tests.

Annex B – Recommendations for Surface Temperature Measurements and Thermocouple Usage

B.1 Temperature Measurement Surface

The temperature on the remote phosphor component, which is most easily measured, is typically the external surface. However, this temperature should be correlated to the phosphor temperature if the manufacturer's maximum operating condition is defined for the phosphor. If the phosphor layer is applied to a thick substrate (either diffusing or clear layer), there can be a substantial thermal gradient from the phosphor layer (where the heat is generated) to the opposite side of the substrate. To minimize the temperature errors caused by this gradient, the maximum surface temperature to be measured is on the surface that is closer to the phosphor layer and can be monitored with a thermocouple inserted into a hole that is deep enough for the thermocouple junction to make contact with this surface (see **Figure C-1**, **Annex C**). The thermocouple could also be in contact with the surface facing the pumping LEDs, but it is important that the thermocouple not absorb light from the pump sources.

B.2 Measurement of Maximum and Average Surface Temperature

The maximum surface temperature is typically located and characterized using an infrared (IR) camera sensitive in the 7- to 11-micron region. In case the phosphor surface temperature is on a material surface that cannot be measured by an IR camera, a thermocouple should be used. The average surface temperature can also be measured by most IR cameras after indicating the excitation area of the remote phosphor.

The emissivity of the IR camera shall be set to the appropriate level for the substrate tested. If there is any doubt concerning the material emissivity, a test is recommended that compares the thermocouple reading to the camera at a fixed DUT temperature (for example, a hot plate at a fixed temperature). This ensures that the emissivity is properly set and the IR camera readings are accurate. Thermocouples can also be used for DUT temperature logging, but extreme care should be taken to minimize heating due to pump light absorption if the thermocouple is positioned between the LEDs and the remote phosphor. This absorbed radiation may result in an abnormally high and inaccurate reading.

B.3 Recommendations for Thermocouple Testing of Remote Phosphor

The remote phosphor source temperature should be maintained below the manufacturer-specified temperature. The DUT temperature increases due to the conversion process of the phosphor. The heating loss in this conversion process depends on the ratio of the converted wavelength energy, hc/λ_c, and the incident excitation energy, hc/λ_e, and is often referred to as the *Stokes' loss*. Additional heat may be generated due to the absorption of spectral output that is not converted by the phosphor and not transmitted through the substrate.

For remote phosphor substrates with low thermal conductivity, heat conduction to the housing assembly can be very minimal. Most of the heat dissipation in this case is by radiation and convection. For installations that limit free convection on one or more sides of the DUT (a sealed mixing chamber, for example), the inside surface of the DUT may effectively experience a substantially higher ambient temperature than indicated by the air temperature measurements of **Section 5.1.2**. The DUT should be carefully tested, with such installations in mind, under the highest applicable power and environmental conditions. Alternatively, a thermocouple can be mounted externally but in very close proximity to the phosphor layer as described in **Annex C**.

Annex C – Thermal Test Recommendations

To monitor the phosphor layer temperature, the use of an insulated K-type thermocouple no smaller than 36 AWG and no larger than 24 AWG is recommended. The thermocouple wire insulation is stripped, if necessary, where lit by the DUT, to minimize absorbing area. The wires are separated from each other. The thermocouple can be bonded to the surface of the DUT by using a high temperature adhesive. It is recommended that the adhesive have high thermal conductivity, and a white or clear material will tend to reduce light power absorption. To reduce the impact of this absorbed power on the thermocouple, it is further recommended that the thermocouple be mounted such that the weld bead is inside or as near as possible to the phosphor layer, as shown in **Figure C-1** for an extruded remote phosphor material.

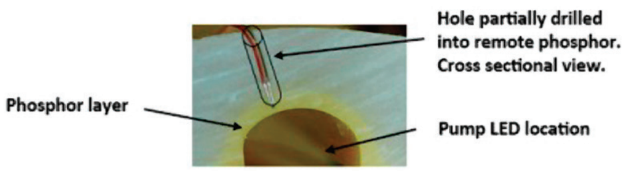

Figure C-1. Recommended thermocouple location.

It is important to note that in **Figure C-1** a small hole matched to the thermocouple diameter is drilled through the outside surface of the substrate, and partially into the remote phosphor layer. For very thin materials it might be necessary to bond to the phosphor-containing surface.

The following steps are used to complete the thermocouple installation for a disk:

Step 1. Drill into non-phosphor-coated surface at least 85% of the thickness of the material using a small flat-bottom drill bit or small engraving tool, shown in **Figure C-2** (recommended tolerance ±0.1 mm for depth).

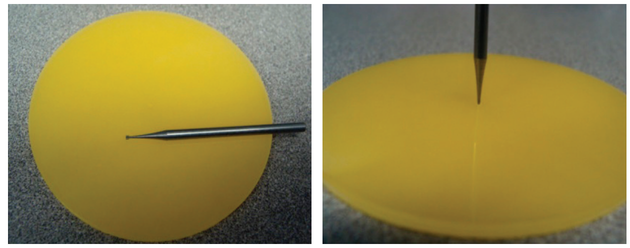

Figure C-2. Drilling into surface.

Step 2, Dispense a small amount of epoxy (recommended service temperature rating of 120 °C or higher) on a clean surface, as shown in **Figure C-3**. If a two-part epoxy is used, it is important to fully mix the materials prior to applying. If the remote phosphor maximum operating temperature is above the epoxy operating temperature, the thermocouple may need to be held in place mechanically using a small Teflon plug or other suitable high-temperature material.

Figure C-3. Dispensing epoxy.

Step 3. Dip the head of the thermocouple weld bead into the premixed epoxy, as shown in **Figure C-4**, or set the thermocouple in place and dab the epoxy onto the thermocouple.

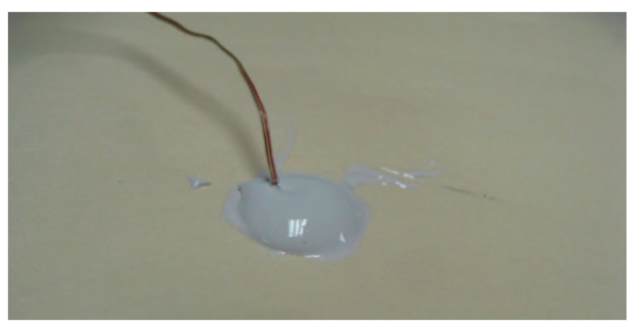

Figure C-4. Dipping weld bead into epoxy.

Step 4. Push the thermocouple bead into the hole on the exterior side of the remote phosphor source, and bend the thermocouple in such a way as to apply a constant push of the thermocouple against the bottom of the drilled hole. Place a small amount electrical tape on the thermocouple to hold it while drying, as shown in **Figure C-5**.

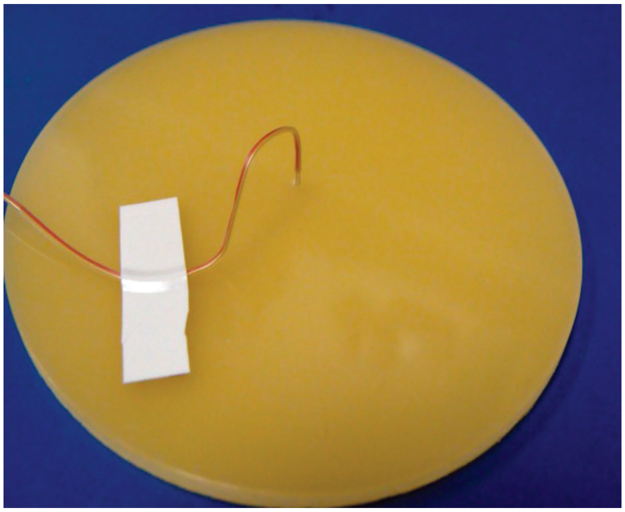

Figure C-5. Tape holding thermocouple in place during drying.

Appropriate drying time should be allowed, while maintaining surface contact between the thermocouple and remote phosphor. Care should be used when moving the assembly, even after the epoxy is dry, since the thermocouple can be pulled away. Taping the thermocouple away from the remote phosphor to provide strain relief during testing is also recommended.

The DUT can then be installed in the luminaire and the thermocouple routed to the appropriate meter for temperature monitoring under the operating environment selected. The tape is removed and the remote phosphor cleaned according to the manufacturer's recommendations.

Measurement of the inside surfaces of the remote phosphor is subject to greater inaccuracies, due to absorption of light by the thermocouple. To reduce this source of error, the temperature can be recorded every two seconds or less, for a period of one to two minutes after turning the pump LEDs off. The temperature data taken after thermocouple stabilization are then used to estimate data trajectory without absorption using a regression fit of the dataset, excluding the initial rapid drop. Comparison of the curve-fit with the measured data reveals the temperature error caused by light absorption in the thermocouple. The difference between the temperature measured at the turn-off time and the trend plot at the turn-off time provides the temperature error due to light absorption. **Figure C-6** shows an example plot using this technique.

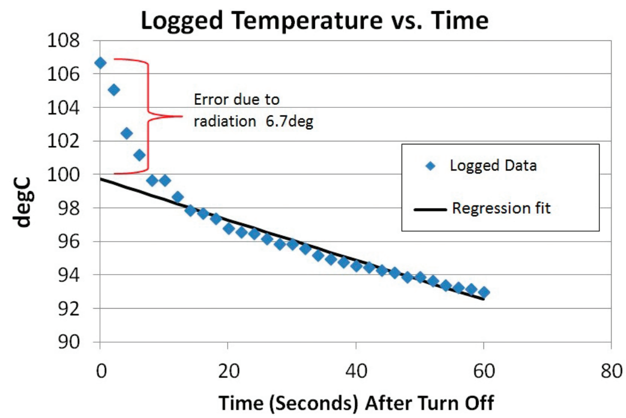

Figure C-6. Absorption correction.

Annex D – Photometric Measurement Recommendations

Total radiant output of the pump LED source irradiance is measured with the DUT removed from the test assembly at the specified forward current. Total luminous flux shall also be measured with the DUT installed into the test assembly at the identical forward current. It is recommended that all photometric and colorimetric values be determined from total spectral radiant flux measurements.

Data recorded at each measurement interval should always be consistent with regard to warm-up time, forward current, orientation, and positioning in the test equipment.

ADDITIONAL READING

National Electrical Manufacturers Association. ANSI C78.377-2015, Electric Lamps – Specifications for the Chromaticity of Solid-State Lighting Products. Washington, DC: American National Standards Institute; 2015.

National Electrical Manufacturers Association. NEMA White Paper LSD-68-2013, Remote Phosphor Integral Lamps and Luminaires. Washington, DC: NEMA; 2013.

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